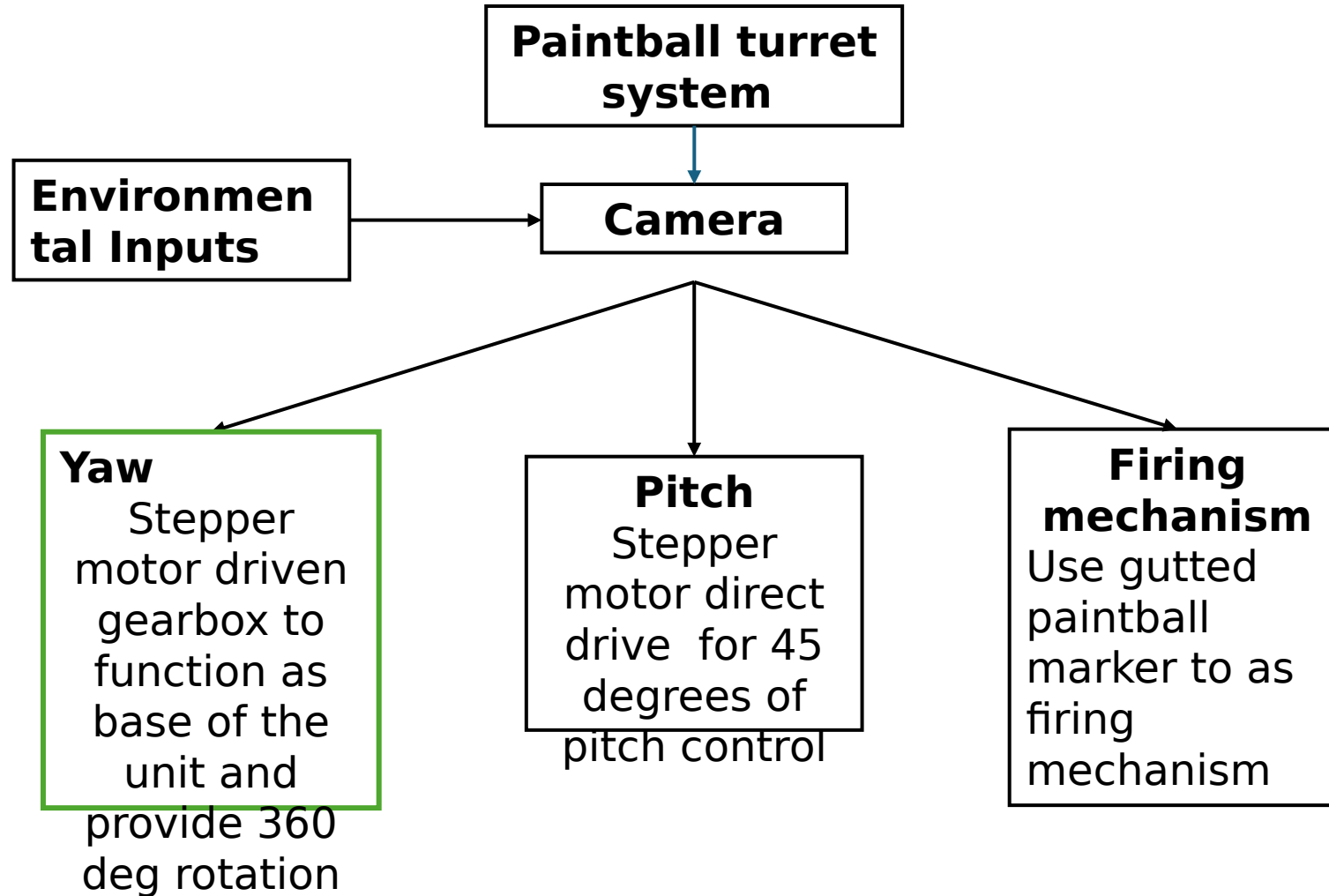


Technical Presentation – Paintball Turret

Harrison Lanfrank –
Mechanical Design
Engineering

Project Overview



Example



Design Targets

Accuracy @ 5m

Metric	Target Value	Notes
Angular Accuracy	±1.0°	Ensures reliable torso-sized hits
Positional Error @ 5m	≤8.7 cm	Equivalent to ±1.0° at 5 meters
Repeatability	Consistent return to set angles	Essential for calibration

Motion Speed

Axis	Target Speed	Use Case
Yaw (Pan)	≥35°/s	Lateral movement of targets @ Jog speed (3 m/s)
Pitch (Tilt)	≥35°/s	Vertical adjustments for elevation

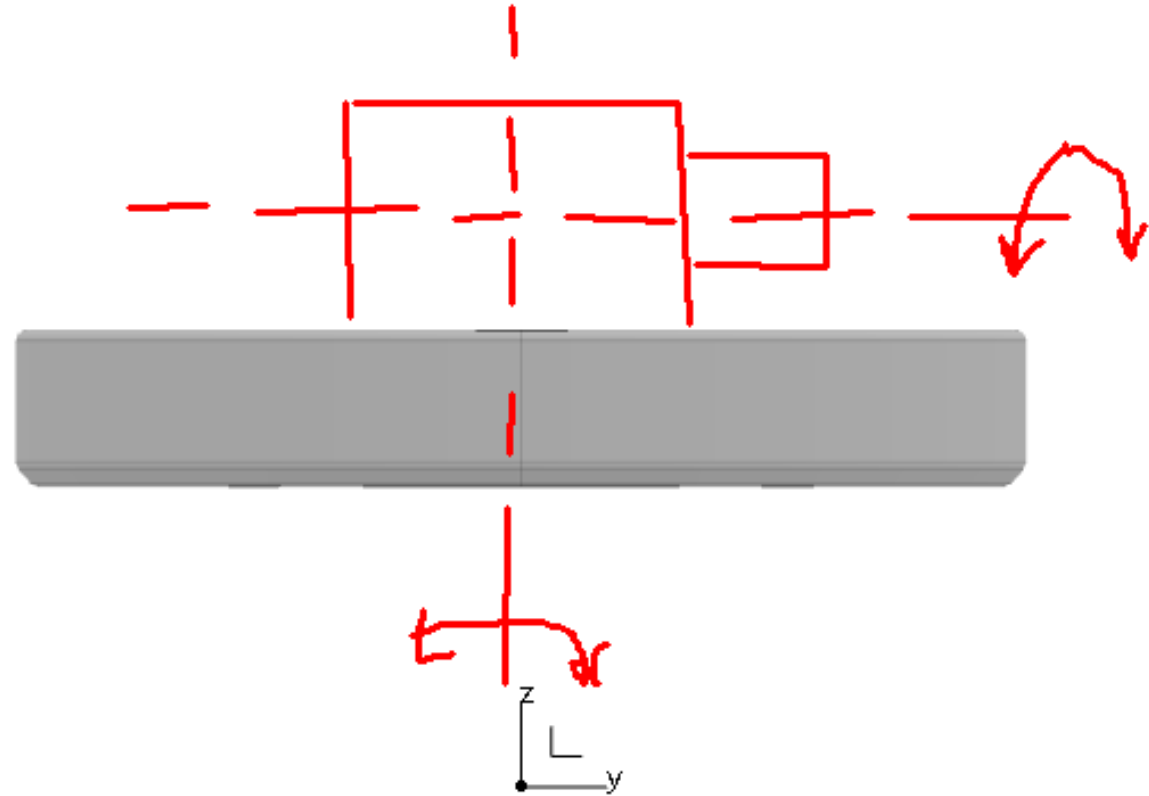
Yaw Mechanism Constraints

Constraints	Target Value	Notes
Cost	<100\$	Prototyping costs for hardware
Packaging Dimensions	100mm Dia x 70mm	Compact packaging
Fabrication	FDM 3D-printing + Amazon	Prioritize for speed of iterations
Output	Rotate 4kg	Weight of pitch mechanism and shaft

Turret Base Design (Yaw)

Concept

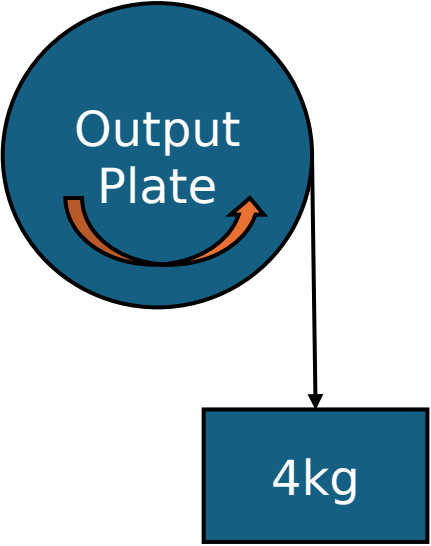
- 360 deg rotating platform to mount pitch and firing mechanisms
- Must reliably rotate 4kg of mass



Turret Base Design (Yaw Function)

Required Torque Output

$4\text{kg} @ 0.05\text{m} = 0.2\text{Nm}$

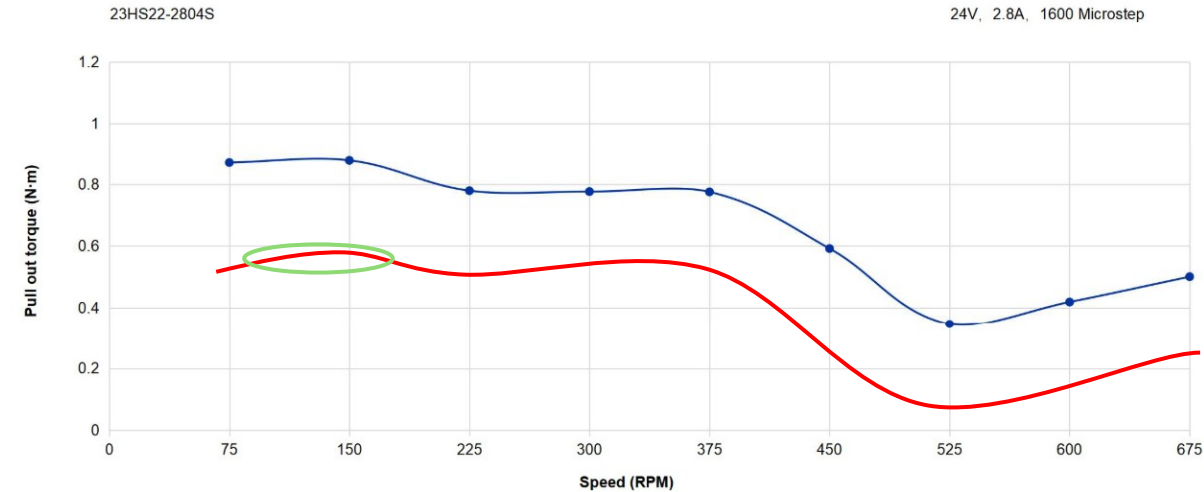


↑
Incorrect

Gear Reduction	15:1	10:1
Required Torque @ Gearbox Output shaft	0.2Nm	0.2Nm
Required Speed @ Output shaft	34.5 deg/s (6 RPM)	34.5 deg/s (6 RPM)
Input Torque	0.013Nm	0.02Nm
Input Speed (RPM)	90	60

*To utilize Nema 23 Stepper I had on hand

12V supply @ 200-800 microstep



Cycloidal Gear

Benefits

- Low backlash
- Minimized friction and wear due to rolling contact
- Compact packaging
- Good for load transfer when using 3D-printed parts
- Common in robotics
- Haven't come across it before

3d-printed spur gears often fail due to concentrated load on small features

Greater contact area between gear and ring pins to distribute load

References

<https://uwaterloo.atlassian.net/wiki/spaces/UWRT/pages/33962263427/Cycloidal+Drive+->

Cycloidal Gear Profile Parametric Equation

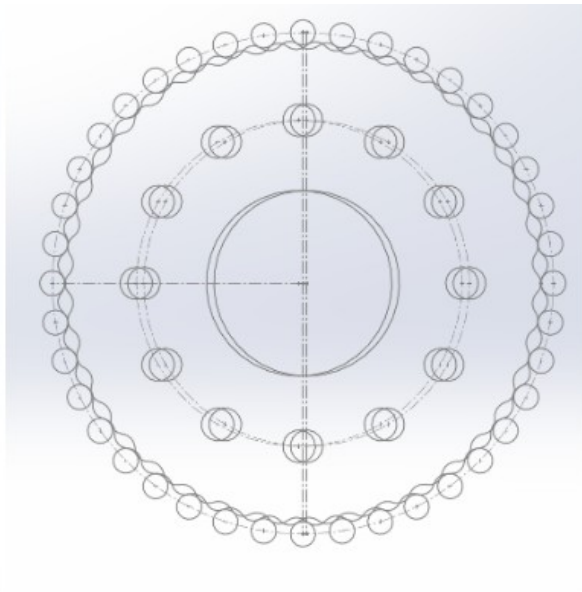
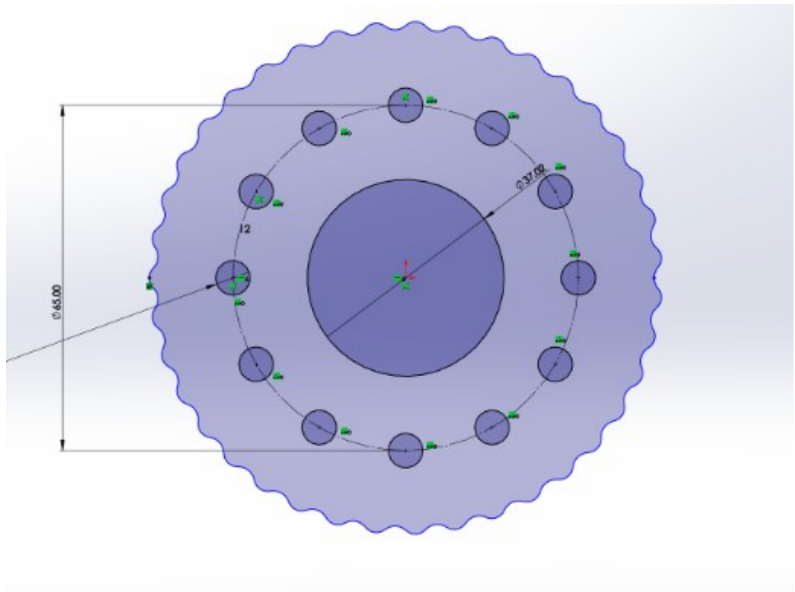
$$X = (R\cos(t)) - (R\cos(t + \arctan(\sin((1-N)t)/((R/EN) - \cos((1-N)t)))) - (E\cos(Nt))$$

$$Y = (-R\sin(t)) + (R\sin(t + \arctan(\sin((1-N)t)/((R/EN) - \cos((1-N)t)))) + (E\sin(Nt))$$

R = Outer lobe Position Radius
Rr= Outer lobe Radius
N = # of lobes (Akin to Reduction
E= Eccentricity

Design Parameters	Value
R	45mm
Rr	6.5mm
N	16
E	1.5mm

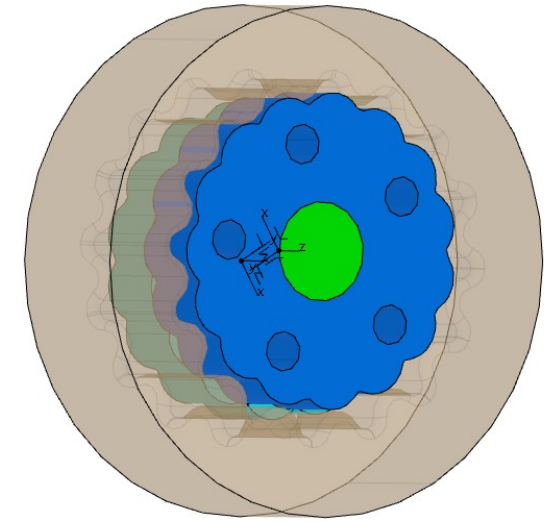
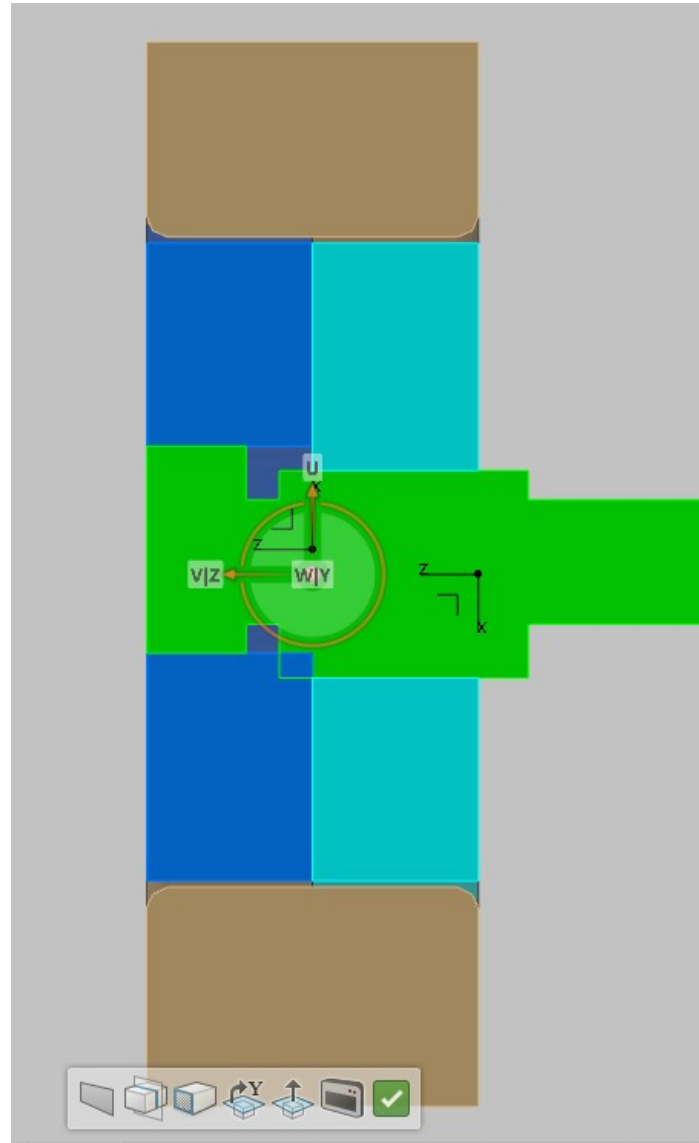
*-1mm offset to account for assembly, and tolerances.



Cycloidal Gear Proof of Concept

DFM Considerations

- FDM printing tolerances $\pm 0.2\text{mm}$
- Design to be compatible with standard hardware
- Limited to PLA
- Limit features that require internal supports
- Print orientation



1st Iteration

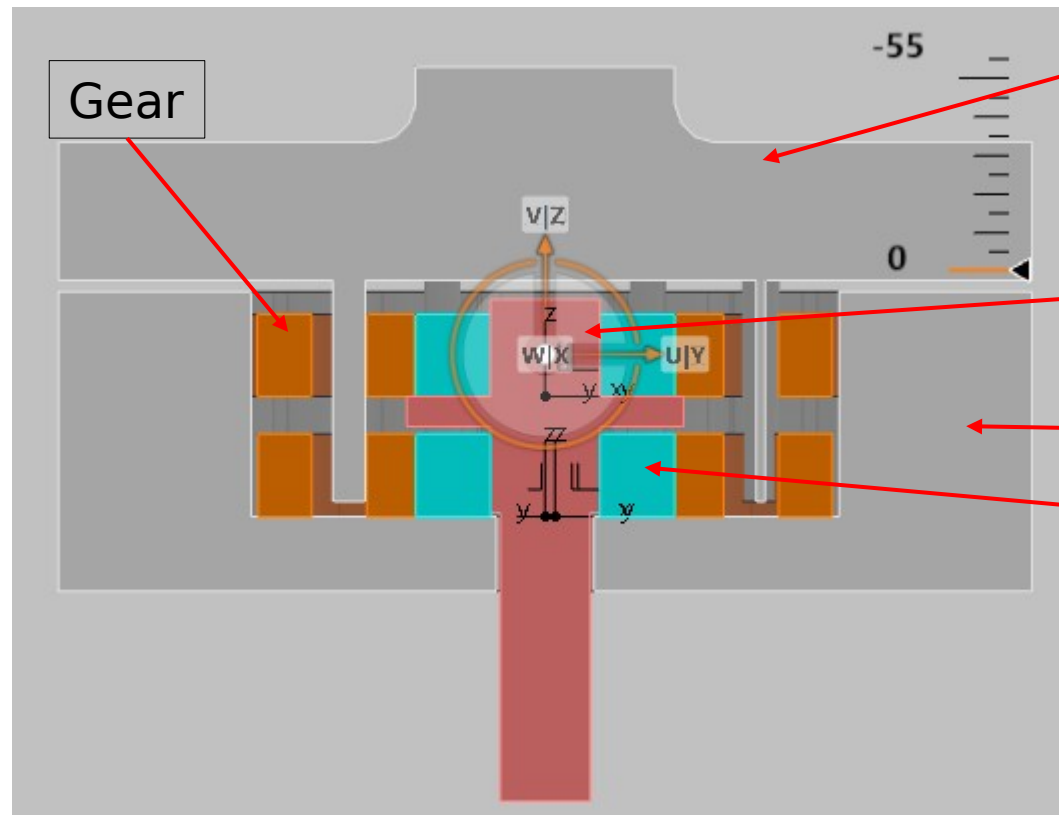
Observations

Shaft

- Lobe 1&2 diameter out of spec (nominal = 14.85, Actual = 14.3-14.5)
- Input shaft broke at shoulder
- High torque to overcome at certain points due to gear getting stuck at some points.
- **Layer lines are in plane with radial forces, so they're loaded in shear and split**

Summary

- works overall to show feasibility
- needs tuning and design improvement
- Holes are 0.2mm undersized



Output Plate

Eccentric Shaft

Hub/outer Bearings

Next Iteration

- Print shaft sideways to prevent layer line aligning with radial load
- Oversize shaft nominal (15.2mm dia)
- Understand printer tolerances in (X,Y,Z, and features)
- Print and test with output shaft



2nd Iteration

Observations

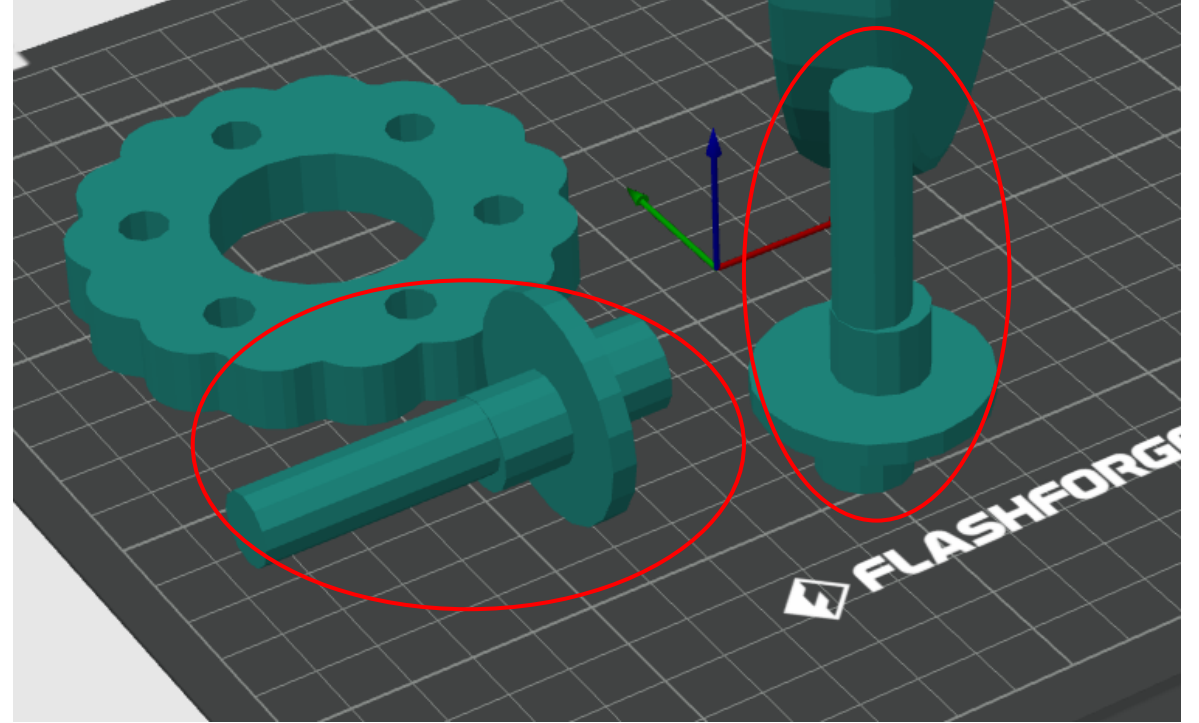
- Oversized lobe diameter to compensate for 3D-print undersizing (14.8 -> 15.2)
- Measured 14.9mm
- Side print had tighter fit due to support layer breakaway on radius of lobe

Summary

- better lobe fitment
- print vertical but oversized diameters to prevent profile distortion due to support layers
- output shaft/plate needs to be print flat
- need to design a feature to add thrust bearing and distribute load effectively
- brainstorm some radial bearing addition to center output plate

Next Steps

- Need a way to measure backlash
- Fix output shaft
- Integrate thrust bearing to transfer axial load
- Integrate radial bearing for output shaft for alignment
- add a feature to make sure output shaft and input shaft are concentric



Fix pins

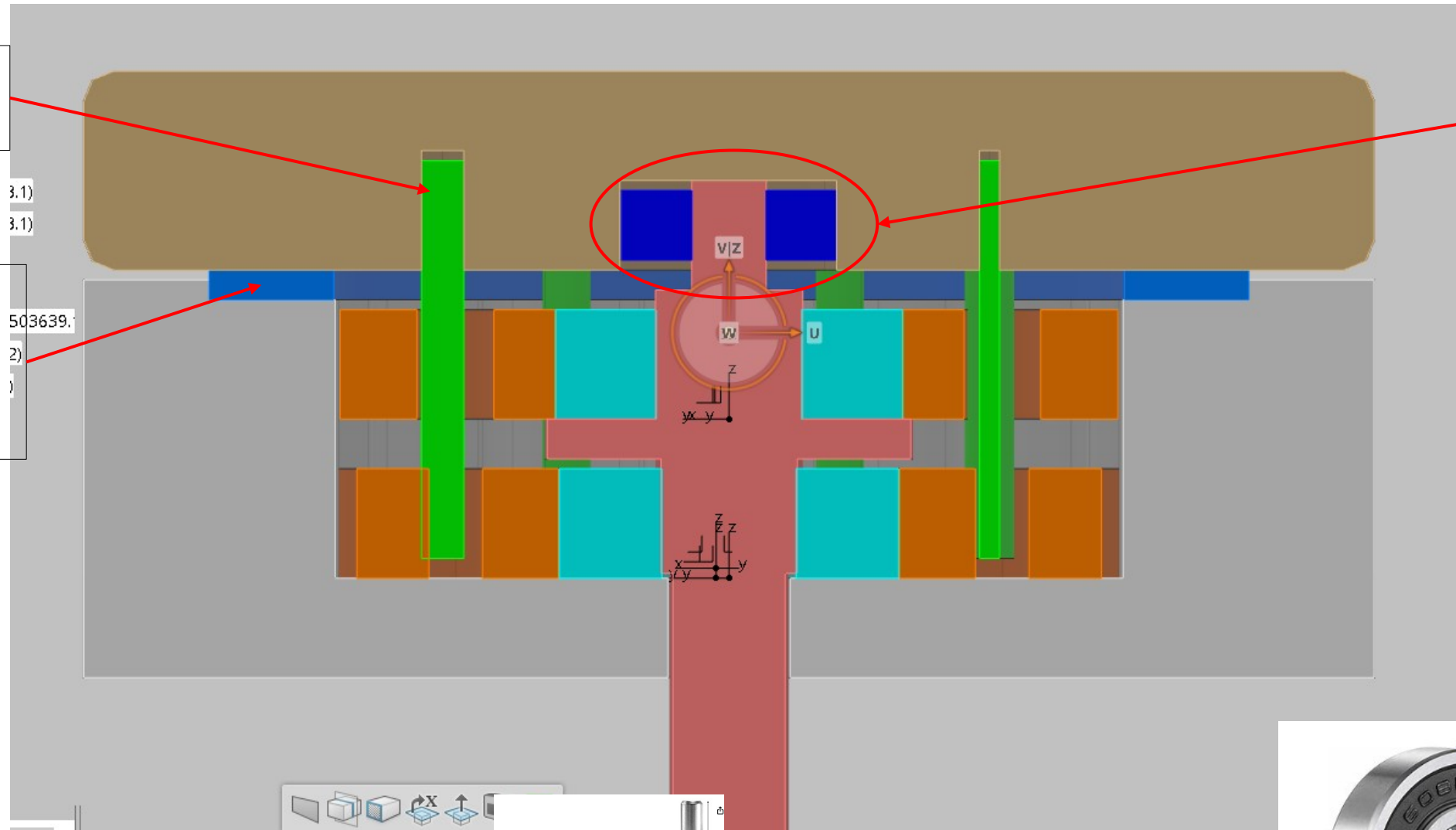
- Print flat
- Press fit for dowel pins?

3rd Iteration

Added
Dowel Pins

Added
thrust
bearing to
reduce
friction

Added
bearing
and feature
on shaft to
align
output
plate



Observations

- Integrated thrust bearing into assembly to handle axial loads from downstream systems
- Integrated additional radial bearing that connects output plate and input shaft concentrically (Still on different axes and can rotate independently)
- **Eccentric shaft is taking all radial load.**
- Sharp shoulders with no stress relief. High chance of failure

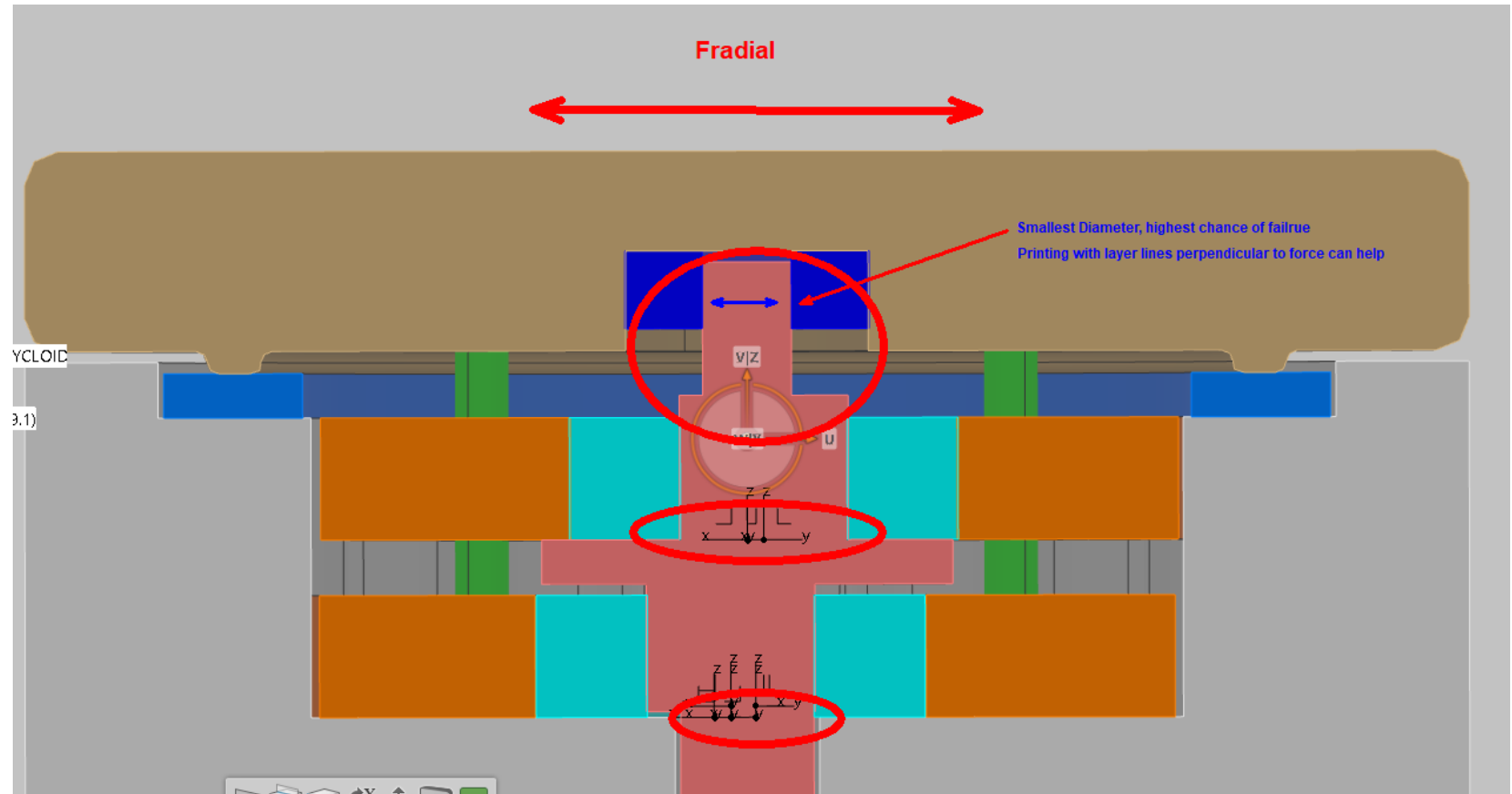
Thrust Bearing Capability

Static Loading = 278.5kN

Dynamic Loading = 55.6kN

Speed = 2600rpm Max

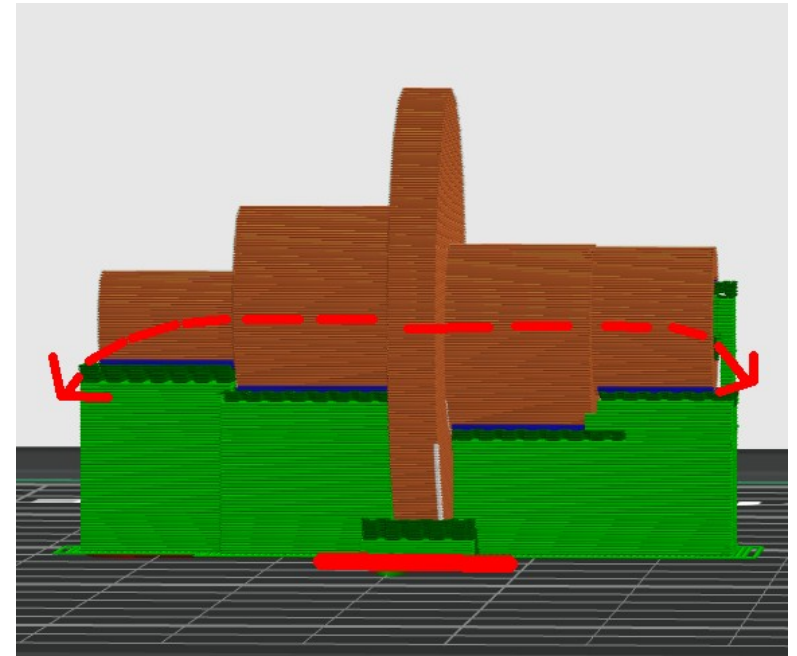
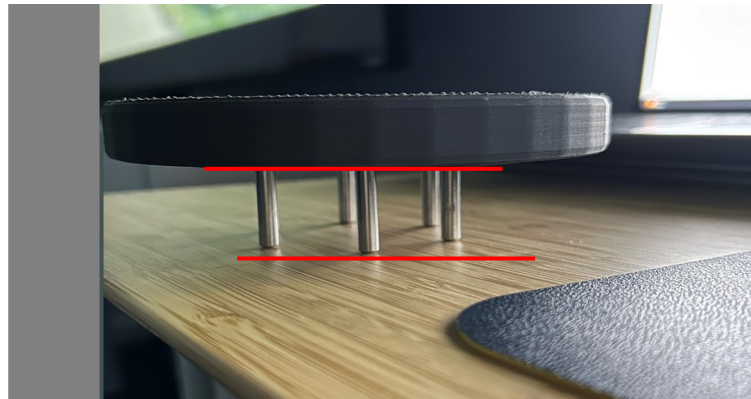
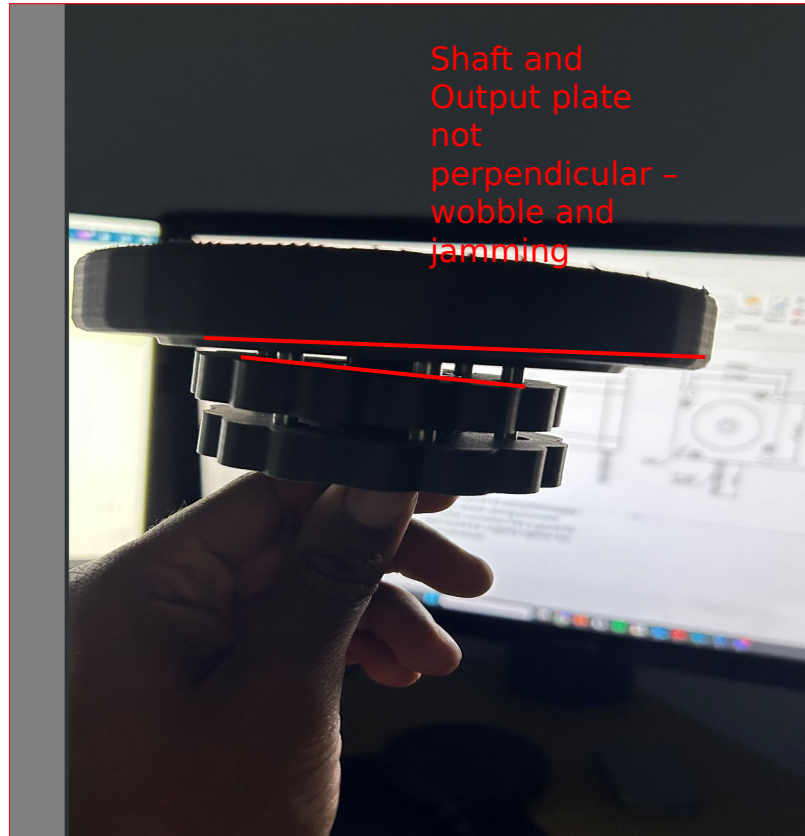
Thrust bearing will not be failure point. Much higher capacity than needed for the system.



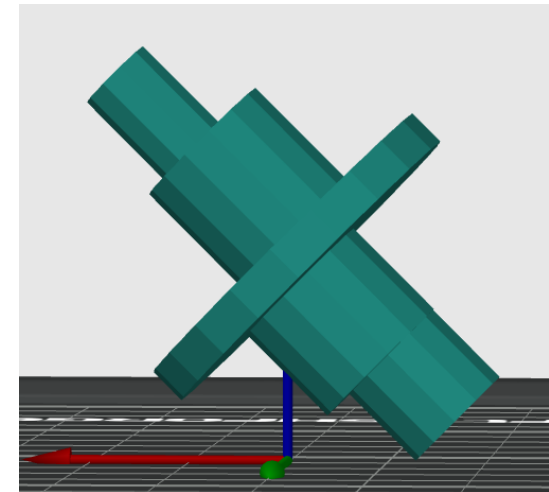
Debugging Assembly Issues

		Nominal Dia	Measured	Variation	Fit
Eccentric Shaft	Input shaft	12	11.8	-0.2	Good Fit
	output shaft	8	7.7	-0.3	Tighter
	lobe 1	15	14.8	-0.2	Good(loose)
	lobe 2	15	14.8	-0.2	Good (loose)
Cycloidal Gear	Bearing Hole	35.2	34.9	-0.3	Perfect
	Pin Holes	8.1	7.8	-0.3	Needs to meet design spec
Hub	Shaft Hole	13	12.7	-0.3	Good
	Thrust Bearing Seat	106	105.4	-0.6	Maybe 0.5mm tighter later
Output Plate	Bearing Hole	22.4	22.2	-0.2	Too tight for prototyping
	Dowel Pin Hole	5.5	5.6	0.1	Needs to be slightly smaller

Debugging Assembly Issues



Eccentric shaft was bowed. Out of perpendicularity along axis. Caused fitment, alignment and uneven loading issues. Occurs during horizontal print orientation



4th Iteration

New Features

- 5mm dowel pins
- 22mm bearing + eccentric shaft aligns concentrically with output plate such that output plate is radially aligned but on a different axis than input
- Thrust Bearing + Output plate touch off ring. Axial loading only through thrust bearing
- Tuned Eccentric shaft (lobe/shaft diameters)

Observations

- Eccentric shaft need tight tolerance.
- Still gets jammed up
- Friction marks on bottom surface of gear due to rubbing with gearbox face



1st Hardware Trial

Observations

- Rattly gearbox and vibration
- Feels like motor is skipping steps (motor only trial)
- Loud gritty sounds from motor

Electronics

- learned that speed is quite limited
- Very jerky and loud at 200 microsteps
- Frequency of missed steps is proportional to increase speed and increase microsteps

Next steps

- Create a test code which takes position and speed as input
- Need to redesign gearbox for smoother operation and reduce points of friction
- **Implement Acceleration function**
- Measure stepper motor accuracy
- Measure gearbox accuracy



Positional Accuracy Testing

Test #1 - Incremental Accuracy at 1 deg steps

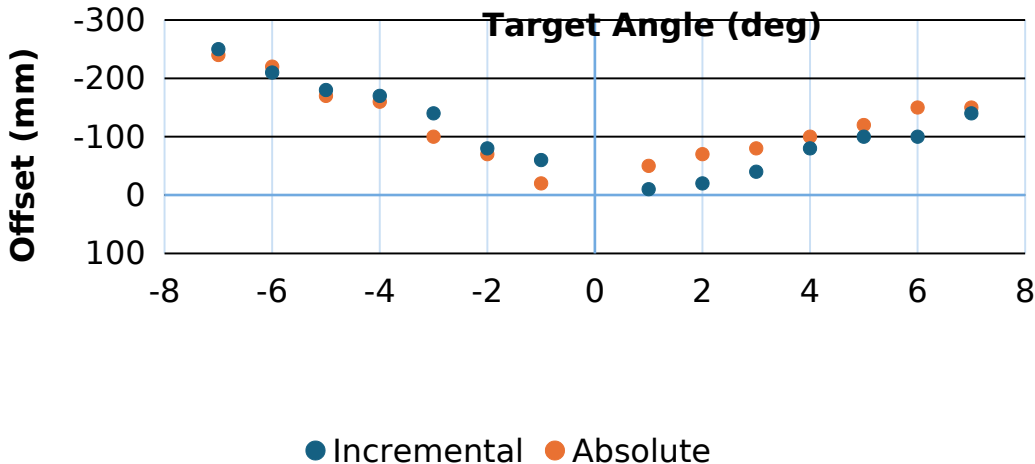
- 1. Measure distance between system and whiteboard.
- 2. Determine correlation between change in angle and change in position (1 deg = 0.08m)
- 3. Set system at position (0,0)
- 4. Provide terminal command to rotate -1 deg in X
- 5. Measure distance between laser and point
- 6. Repeat steps 4-5 till -7
- 7. Manually reset to (0,0)
- 8. Repeat steps 4-5 till +7

Test #2 - Absolute Accuracy

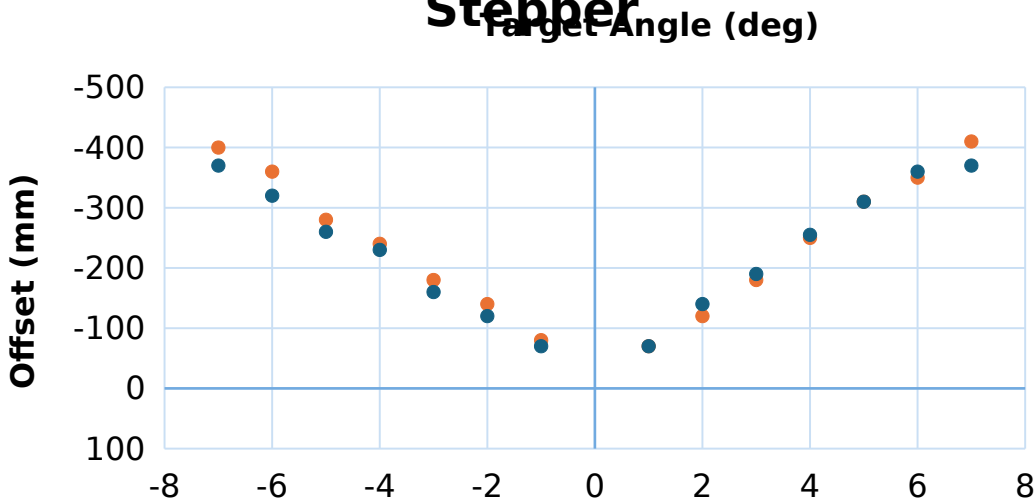
- 1. Start at (0,0)
- 2. Command to -7
- 3. Measure distance and then manually reset to (0,0)
- 4. Command to -6
- 5. Repeat for -7 to +7 (reset in between each position command)

Positional Accuracy Gearbox V1

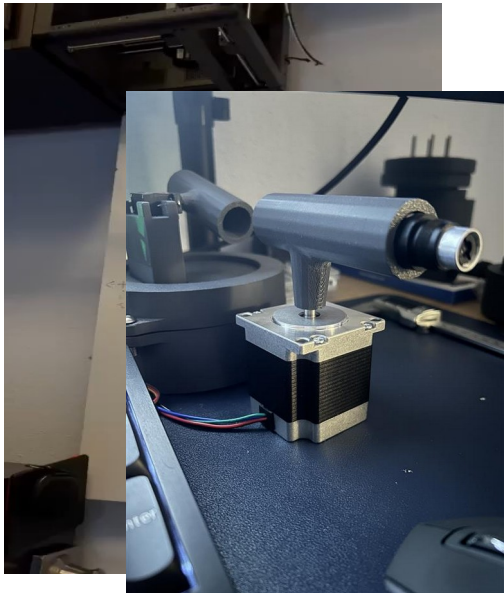
1 deg = 80mm @ 4.3M



Positional Accuracy - Direct to Stepper



Test Parameters	
Materials	PLA to PLA Interface
Microsteps	800
Power	12V, 2.8A
Speed	400 microseconds
Accel	0.3*Speed



V2 Design Improvements

Corrected Torque Calc

Required Input Torque (Motor)

Output Torque

$T = I\alpha$ * Assume mass on top of output plate is uniformly distributed

Solid Disk $\frac{1}{2}mr^2 = I$

$\alpha = 1.22 \text{ rad/s}^2$

$m = 4 \text{ kg}$ ← (Pitch mechanism + Paintball gun)

$r = 0.45 \text{ m}$

$T_{out} = \frac{1}{2}(4)(0.45)^2 \times (1.22)$

$T_{out} = 0.49 \text{ Nm}$

$T_{in} = \frac{T_{out}}{15} \approx 0.03 \text{ Nm}$

Assume $\eta = 50\%$ for gearbox & S.F. of 3

$T_{in,actual} = \frac{T_{in}}{\eta} \times \text{S.F.}$

$T_{in,actual} \approx 0.2 \text{ Nm}$

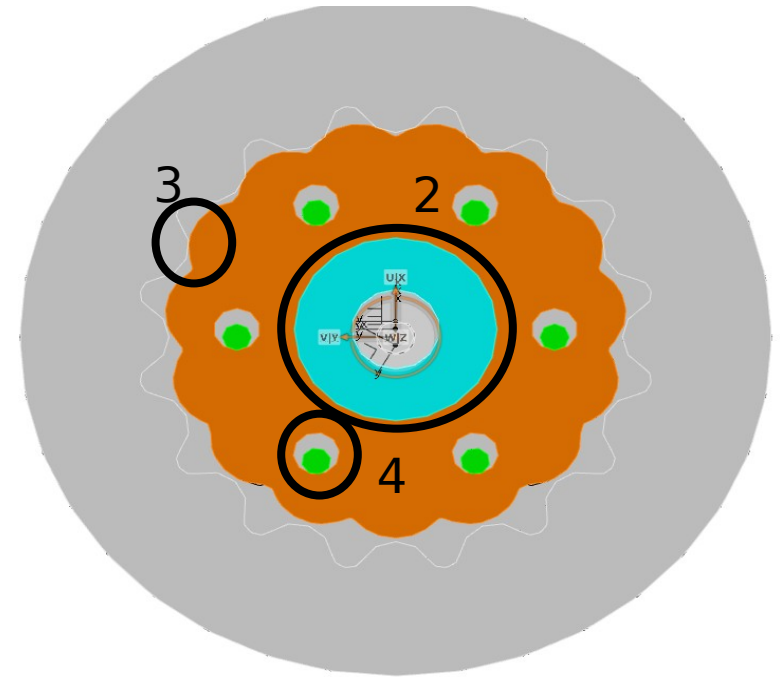
Mechanical Losses

1. Sliding friction between eccentric shaft and gearbox (not shown)
2. Bearing friction between cam and gear
3. Rolling + sliding friction between disk gear and ring gear
4. Sliding friction between output pins and output holes on disk gear

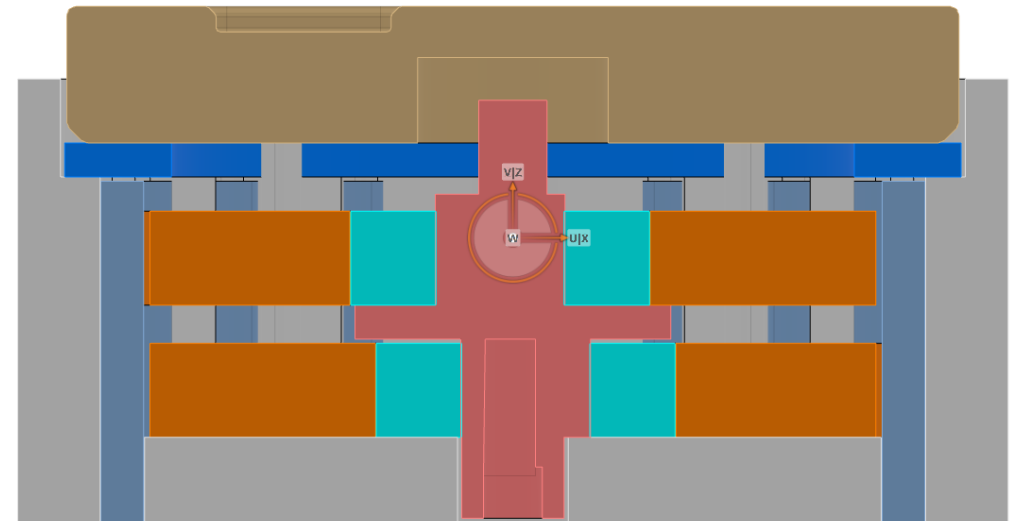
Hardware Improvements

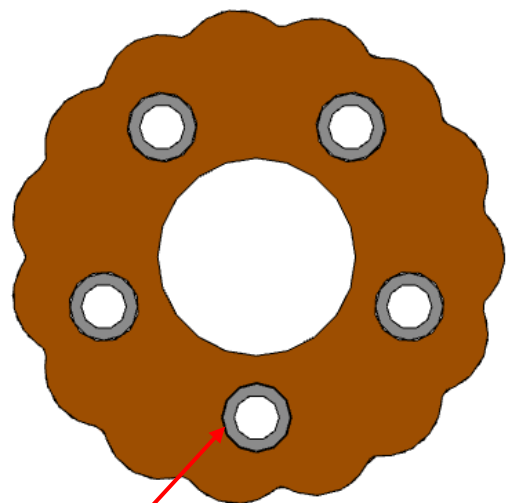
1. Reducing friction in gearbox will minimize missteps from motor
2. Tune micro step & amperage setting (**Driver**)
3. Tune speed and accel parameters (**Software**)

V1

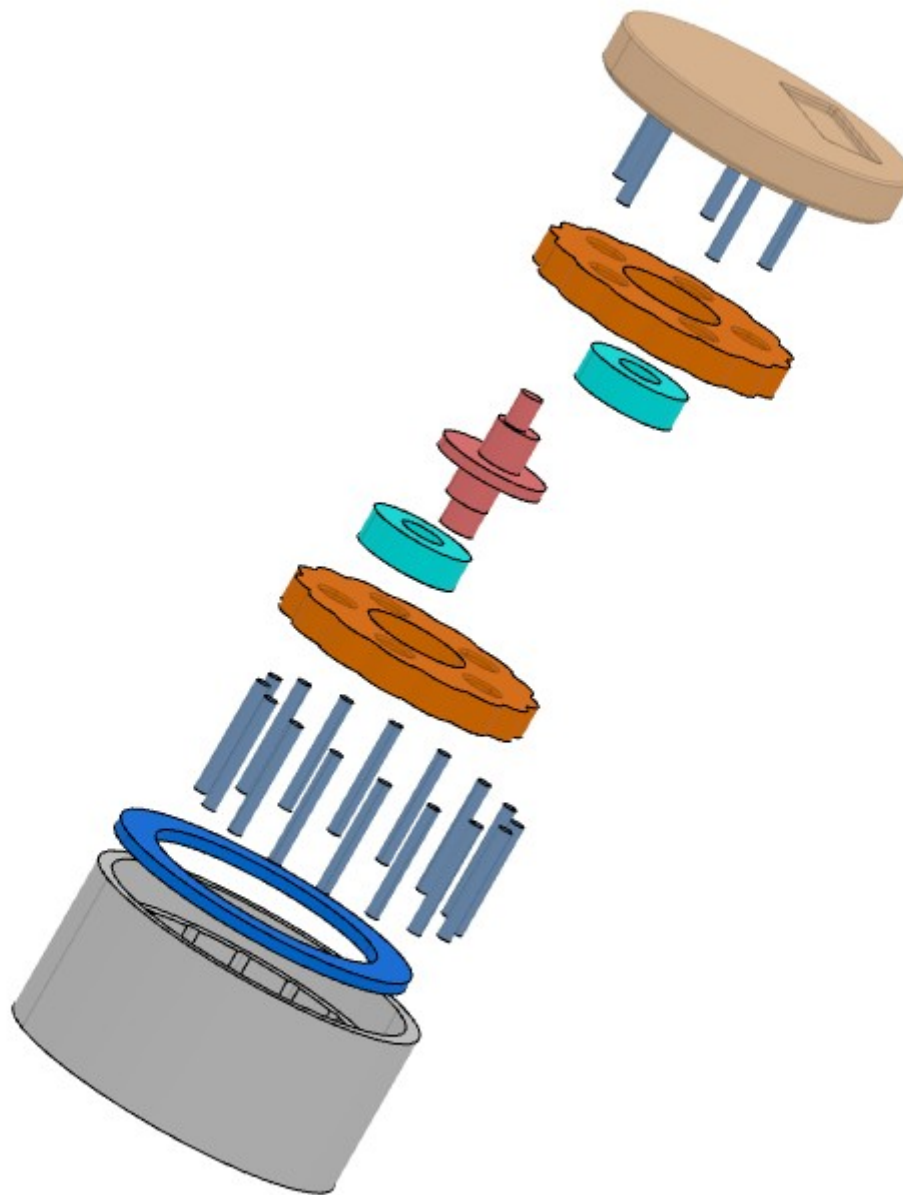


V2

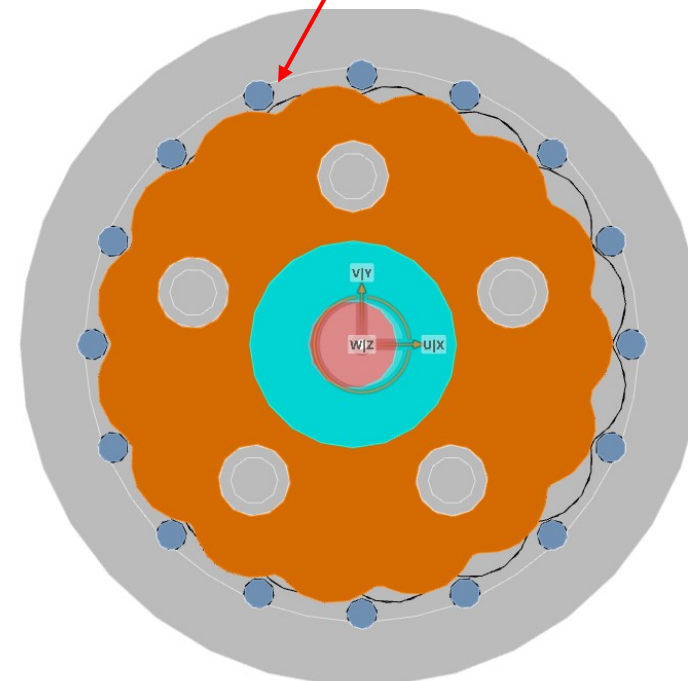




Bearing
for output
pins



Steel dowel pins
to plastic -
rolling + sliding
interface



Learnings

- Motor needs to be loaded for optimal performance. (Reduce resonance)
- Motor struggles for accuracy and speed at small increments of rotation. (Increase gear ratio to allow for speed ramp up to optimal speed thus more accurate positioning)
- Need to offset gear surface since the “theoretical profile” does not account for real world conditions (tolerances, friction, contact stress, and etc.)
- Print orientation matters for managing failure under load
- Helped to prototype and iterate quickly. Was able to have constant improvement.
- Underestimated impact of tolerances on this design, from assembly to actual gear motion
- Motors have many parameters that need to be optimized per application. Not plug and play

Next Steps

1. Assemble V2 design and repeat position accuracy test
2. Add feature to hub design to stabilize the gearbox radially. (more mass or clamp to table)
3. Replace PLA
4. Design test rig to accurately determine torque
5. Torque test - use torque wrench to measure torque at gearbox output and motor input
6. Continue optimizing software and driver to minimize missteps
7. Add lubrication (Vaseline)
8. Investigate operating temp of gearbox

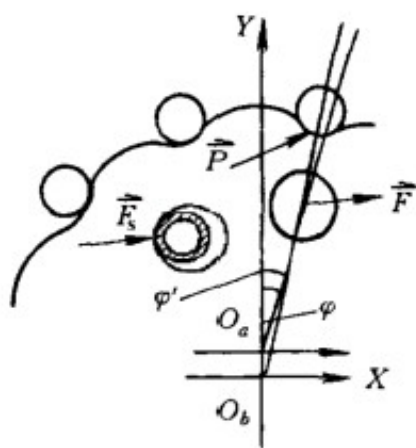
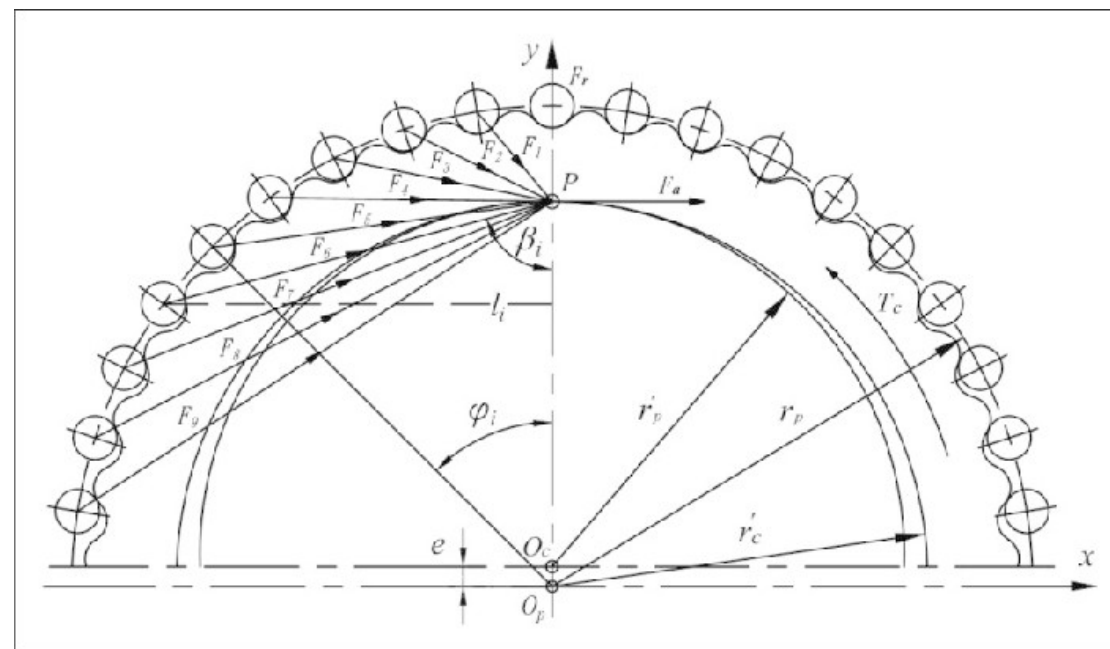


Figure 23. Forces (Yao & Zhang , 1997)



Pressure Angle

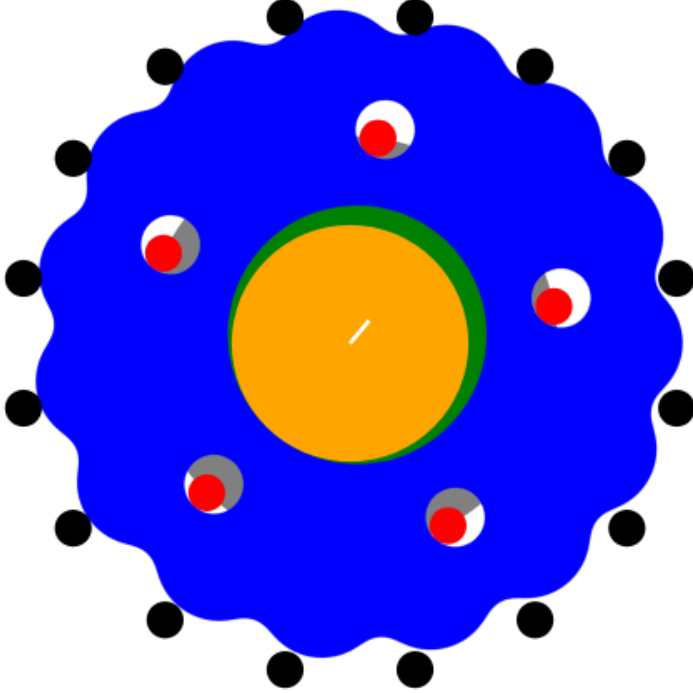
Average pressure angle = 35 deg

*Not an accurate characterization

Pressure angle is proportional to power transmission

Levers for Pressure Angle

- # of external pins a.k.a GR (+)
- Eccentricity (-)
- Pitch circle diameter (+)
- Ring pin diameter (-)



Input: Eccentric Shaft
Output: Pin disk

Transmission Ratio: 15 ?

Output speed: 3.267 ?

Average Pressure Angle: 35.00 ?

Range: 0.27 ?

Download Cycloid Profile (DXF format) ?

Input speed:	49	?	Fixed Ring Diameter:	90	?
Number of External Pins:	16	?	External Pin Diameter:	5	?
Number of Output Pins:	5	?	Output Pin Diameter:	5	?
Output Disk Diameter:	56	?	Number of lobes:	15	?
Eccentricity:	1.5	?	Camshaft Hole:	35.00	?

PLA vs Nylon

Material Properties	PLA	Nylon
Yield Stress (MPa)	38	77
Youngs Modulus (GPa)	3.5	2.6
Elongation @ yield (%)	6	46
Coefficient of Friction	0.42	0.26
Hardness (Shore D)	66	81
	PLA	Nylon
Cost	\$	\$\$\$
Difficulty to print	Easy	Moderate (need heated enclosure for optimal results)

“Zero Backlash”

Traditional Involute Gear Profile

- Gap between teeth are part of the design to prevent jamming due to manufacturing tolerances
- Load concentrated on a few teeth and has sliding and rolling contact

Cycloidal Gear Profile

- Supposed to have constant contact (no design gap). Will be in preload
- Multi-tooth contact – up to 50% of pins
- When reversed the load transfers onto the pins already in contact

Force on Output Pins

$$F_{smax} = \frac{4.8 T_g}{Z_w r_2}$$

$$T_g = \frac{T_{output}}{\# \text{ of disks}}$$

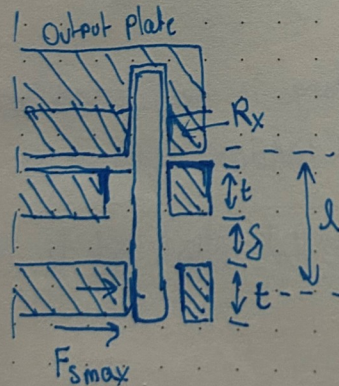
Assume ideal Toutput

$$T_g = \frac{0.5 \text{ Nm}}{2} = 0.25 \text{ Nm}$$

$$F_{smax} = \frac{4.8 (0.25)}{(5) (0.28)}$$

$$F_{smax} = 0.86 \text{ N}$$

Bending stress on Pin



Stainless steel
 $\sigma_y = 215 \text{ MPa}$

ultimate shear = $(0.6) (UTS)$
 $= 303 \text{ MPa}$

T_g = designated torque per disc

Z_w = # of output pins

r_2 = radius of pin circle

$$\sigma_b = \frac{M y}{I}$$

$$I = \frac{\pi d_p^4}{32} = 1.4 \times 10^{-11} \text{ m}^4$$

$$l = t + 0.5t + s$$

$$l = 0.011 \text{ m}$$

$$M = F_{smax} \times l = 0.018 \text{ Nm}$$

$$\sigma_b = 44.6 \text{ MPa}$$

$$\sigma_b = 1.47 \text{ MPa}$$

$\sigma_b \gg \sigma_{bmax}$ No failure

$R_x = 0.86 \text{ N}$ on plastic gear

